The documentation and process conversion measures necessary to comply with this revision shall be completed by 25 May 2016.

INCH–POUND

MIL–PRF–19500/358G
25 January 2016
SUPERSEDING
MIL–PRF–19500/358F
12 July 2013

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, VOLTAGE REGULATOR,
STUD MOUNT PACKAGE, STANDARD AND REVERSE POLARITY, TYPES 1N3305B THROUGH 1N3350B AND 1N4549B THROUGH 1N4554B, QUALITY LEVELS JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL–PRF–19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for 50 watt silicon voltage regulator diodes in both standard polarity and reverse polarity. Four levels of product assurance (JAN, JANTX, JANTXV, and JANS) are provided for each device type as specified in MIL–PRF–19500.

1.2 Package outline. The device package outline is a modified package consisting of a DO–5 base with a DO–4 cap in accordance with figure 1 for all encapsulated device types.

1.3 Maximum ratings. Unless otherwise specified T_C = 25ºC. The maximum ratings are as shown in columns 3, 7, and 9 of the characteristics and test ratings table, 3.7, and as follows:

a. Derate P_T = 50W at $T_C \geq$ +75ºC at 0.5 W/ºC above $T_A \geq$ +75ºC.

b. –65ºC ≤ T_J ≤ +175ºC; –65ºC ≤ T_{STG} ≤ +175ºC.

c. Thermal resistance ($R_{\theta JC}$) = 2.0ºC/W maximum.

1.4 Primary electrical characteristics. The primary electrical characteristics are as shown in columns 1, 8, 11, and 12 of the characteristics and ratings table, and 3.7 herein.

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL–PRF–19500 and as specified herein.

1.5.1 JAN certification mark and quality level. The quality level designator, if applicable, is in accordance with MIL–PRF–19500. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", "JANTXV", and "JANS".

1.5.2 Device type. The designation system for the devices covered by this specification sheet is as follows.

1.5.2.1 First number and first letter symbols. The devices of this specification sheet use the first number and letter symbols "1N".

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1.5.2.2 Second number symbols. The second number symbols for the devices covered by this specification sheet are as follows:

3305	3311	3317	3323	3329	3335	3341	3347	4549
3306	3312	3318	3324	3330	3336	3342	3348	4550
3307	3313	3319	3325	3331	3337	3343	3349	4551
3308	3314	3320	3326	3332	3338	3344	3350	4552
3309	3315	3321	3327	3333	3339	3345		4553
3310	3316	3322	3328	3334	3340	3346		4554

1.5.3 Suffix symbols. The following suffix letters are incorporated in the PIN for this specification sheet. All devices covered by this specification sheet use a suffix symbol or symbols as follows.

B	Indicates a modified version of the diode which have a nominal voltage tolerance of ±5 percent over the basic numbered (non-suffix) device.
RB	Indicates reverse polarity packaging of the modified version of the diode.

1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS–19500.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL–PRF–19500 – Semiconductor Devices, General Specification for.

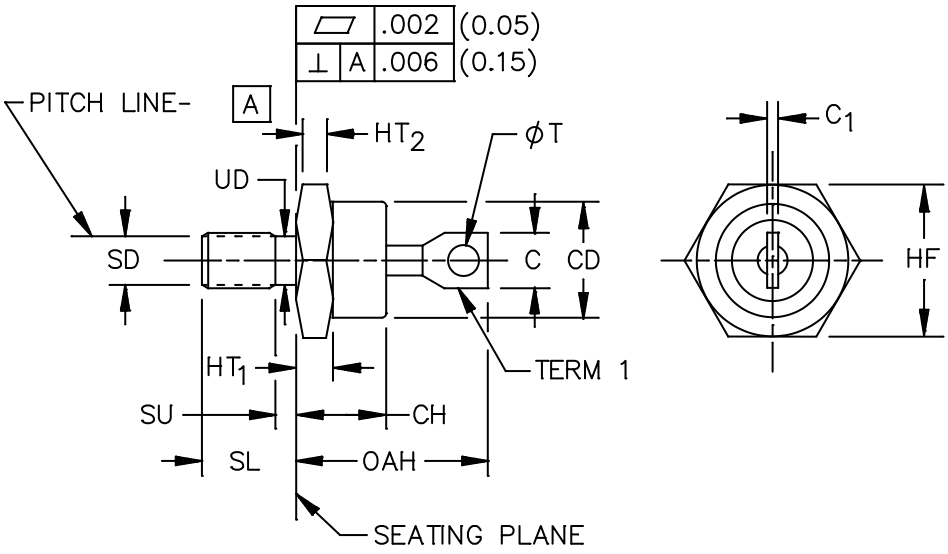
DEPARTMENT OF DEFENSE STANDARDS

MIL–STD–750 – Test Methods for Semiconductor Devices.

(Copies of these documents are available online at http://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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modified DO–5 with a DO–4 cap

Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
C		.250		6.35	
C1	.012	.065	0.305	1.65	3
CD	.255	.424	6.48	10.77	
CH		.450		11.43	
HF	.667	.687	16.94	17.45	
HT_1	.115	.200	2.92	5.08	
HT_2	.060		1.52		
OAH		1.000		25.4	
SD	.250-28				
SL	.422	.453	10.72	11.51	
SU					4
UD	.220	.249	5.59	6.32	
ΦT		.175		4.45	

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Devices with "B" suffix have the anode connected to the case and devices with "RB" suffix (reverse polarity) have the anode connected to the terminal.
3. Angular orientation of this terminal is undefined.
4. Complete threads to extend to within 2.5 threads of seating plane.
5. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.

FIGURE 1. Physical dimensions (modified DO–5 with a DO–4 cap).

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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL–PRF–19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL–PRF–19500 and as follows:

I_{ZM}	Maximum zener current.
I_{ZSM}	Maximum zener surge current.
I_{ZT}	Zener test current.
$V_{Z(reg)}$	Voltage regulation.
Z_Z	Regulator impedance.
Z_K	Knee impedance.
α_{VZ}	Temperature coefficient of regulator voltage.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL–PRF–19500 and herein. Current density of internal conductors shall be as specified in MIL–PRF–19500.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL–PRF–19500, MIL–STD–750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Polarity. Standard polarity devices (B suffix) shall have the anode connected to the case stud. Reverse polarity devices (RB suffix) shall have the cathode connected to the case stud.

3.5 Marking. Marking shall be in accordance with MIL–PRF–19500. The PIN shall be in accordance with 1.5.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.7 Maximum and primary test ratings. The maximum and primary test ratings for voltage regulator diodes shall be as specified in table IV herein.

3.8 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.9 Workmanship. Devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4, and tables I, II, and III).

4.1.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL–PRF–19500, and as specified herein. Lot accumulation period shall be 6 months in lieu of 6 weeks.

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4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL–PRF–19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.3 Screening (quality levels JANTX, JANTXV, and JANS only). Screening shall be in accordance with table E–IV of MIL–PRF–19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen	Measurement	
	Quality level JANS	Quality levels JANTX and JANTXV
3c (1)	Thermal impedance, see 4.3.2.	Thermal impedance, see 4.3.2.
9	I_{R1} and V_Z (for devices with $V_{Z(nom)} \geq 10$ V dc; see column 1 of table IV).	Not applicable.
11	I_{R1} and V_Z; ΔI_{R1} = 100 percent of initial value or 2 μA dc, whichever is greater; ΔV_Z = ±1 percent of initial value (for devices with $V_{Z(nom)} \geq 10$ V dc; see column 1 of table IV).	I_{R1} and V_Z
12	See 4.3.1.	See 4.3.1.
13	Subgroup 2 (except forward voltage test) and subgroup 3 of table I herein; ΔI_{R1} = 100 percent of initial value or 2 μA dc, whichever is greater, ΔV_Z = ±1 percent of initial value.	Subgroup 2 (except forward voltage test) of table I herein; ΔI_{R1} = 100 percent of initial value or 2 μA dc, whichever is greater, ΔV_Z = ±1 percent of initial value.

(1) This test shall be performed anytime after screen 3.

4.3.1 Power burn-in conditions. Power burn-in conditions shall be as follows: $I_Z = I_{ZT}$ (see column 4 of table IV) at T_J = +150ºC minimum.

4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3101 or 4081 of MIL–STD–750, as applicable, using the guidelines in that method for determining I_M, I_H, t_H, t_{SW} (V_C and V_H where appropriate).

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL–PRF–19500 and as specified herein. Group A inspection shall be performed on each sublot.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL–PRF–19500 and table I herein. End-point electrical measurements shall be in accordance with table I, subgroup 2 herein.

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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E–VIA (for quality level JANS) or table E–VIB (for quality levels JAN, JANTX, and JANTXV) of MIL–PRF–19500 and herein. Delta electrical measurements for quality level JANS shall be in accordance with table II herein.

4.4.2.1 Quality level JANS (see table E-VIA of MIL–PRF–19500).

Subgroup	Method	Condition
B4	1037	2,000 cycles. See 4.5.1.
B5	1027	I_Z = column 4 (I_{ZT}) of table IV for 168 hours; T_A = 125ºC or adjusted, as required, to give an average lot; T_J = +225ºC. Option 1 – T_J = +200ºC, 336 hours. Option 2 – T_J = +175ºC, 1,000 hours.
B6	4081	$R_{\theta JC}$ = 2.0ºC/W maximum. For purposes of this test "junction to case" shall be used in lieu of "junction to lead" and $R_{\theta JC}$ shall be used in lieu of $R_{\theta JL}$. The case shall be the reference point for calculation of junction to case thermal resistance ($R_{\theta JC}$). The mounting arrangement shall be with heat sink to case.

4.4.2.2 Quality levels JAN, JANTX and JANTXV (see table E-VIB of MIL–PRF–19500).

Subgroup	Method	Condition
B2	4066	I_{ZSM} = column 9 of table IV.
B3	1027	I_{ZT} = column 4 of table IV, adjust T_A, mounting, or both to achieve T_J = +150ºC minimum.
B3	1037	2,000 cycles. See 4.5.1.
B5	4081	$R_{\theta JC}$ = 2.0ºC/W maximum. For purposes of this test "junction to case" shall be used in lieu of "junction to lead" and $R_{\theta JC}$ shall be used in lieu of $R_{\theta JL}$. The case shall be the reference point for calculation of junction to case thermal resistance ($R_{\theta JC}$). The mounting arrangement shall be with heat sink to case.

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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E–VII of MIL–PRF–19500 and as follows. Delta electrical measurements shall be in accordance with table II herein.

Subgroup	Method	Condition
C2	2036	Tension: Test condition A, 20 pounds (9.07 Kg), t = 15 seconds;
C2	2036	Torque (stud): Test condition D2, 30 lb-in (3.39 newton meter), t = 30 seconds;
C2	2036	Bending stress: Test condition F, method B; 3 pounds (1.36 Kg), t = 15 seconds.
C5		Not applicable.
C6	1027	I_{ZT} = column 4 of table IV, adjust T_A, mounting, or both to achieve T_J = +150ºC minimum.
C6	1037	6,000 cycles. See 4.5.1.
C8	4071	Temperature coefficient of regulator voltage (see 4.5.5) I_{ZT} = column 4 of table IV; T_{ref} = 25ºC±3ºC; T_{test} = T_{ref} +100ºC each sublot; α_{VZ} (column 12 of table IV) = percent/°C; the sample plan shall be n = 22, c = 0; For small lot, the sample plan shall be n = 12 devices, c = 0.
C9		Voltage regulation (see 4.5.6), each sublot, the sample plan shall be n = 22, c = 0.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows. Unless otherwise specified herein, all inspections shall be made at a T_C of 25°C ±3°C.

4.5.1 DC intermittent operation life. The DC intermittent operation life test shall be performed in accordance with test method 1037 of MIL–STD–750, except that the procedure shall be as follows: A cycle shall consist of an "on" period, when forward current is applied suddenly, not gradually, to the device for the time necessary to achieve an increase (delta) case temperature of +85ºC +15ºC, –5ºC followed by an "off" period, when the current is suddenly removed for cooling the case through a similar delta temperature. Auxiliary (forced) cooling is permitted during the "off" period only. Forward current and "on" time, within specific limits, and "off" time may be adjusted to achieve the delta case temperature. Heat sinks shall only be used, if and to the degree necessary, to maintain test samples within the desired delta temperature tolerance. The heating time shall be such that 30 s ≤ $t_{heating}$ ≤ 180 s. The forward current may be steady-state dc, full-wave rectified dc, or the equivalent half-sine wave dc of the specified value. Alternately, I_{ZT} may be used to achieve heating. The test duration shall be the specified number of cycles specified (see 4.4.2.1, 4.4.2.2, and 4.4.3). Within the time interval of 50 cycles before and 500 cycles after the termination of the test, the sample units shall be removed from the specified test conditions and allowed to reach room ambient conditions. Specified end-point measurements for qualification and conformance inspections shall be completed within 96 hours after removal of sample units from the specified test conditions. Additional readings may be taken at the discretion of the manufacturer.

4.5.2 Maximum zener surge current (I_{ZSM}). The maximum zener surge current test shall be performed in accordance with condition B of test method 4066 of MIL–STD–750. I_{ZSM} (see column 9 of table IV) shall be applied in the reverse direction and shall be superimposed on I_{ZT} (see column 4 of table IV) a total of five surges at 1 minute intervals. Each individual surge shall be a one-half square wave pulse of 1/120 second duration or a one-half sine wave with the same effective (rms) current.

4.5.3 Regulator voltage (V_Z). The regulator voltage test shall be performed in accordance with test method 4022 of MIL–STD–750. I_{ZT} (see column 4 of table IV) shall be applied until thermal equilibrium is obtained.

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4.5.4 Reverse current (I_R). The reverse current leakage test shall be performed in accordance with the DC method of test method 4016 of MIL–STD–750. The specified reverse voltage shall be applied to the terminals and the reverse current measured.

4.5.5 Temperature coefficient of regulator voltage (α_{VZ}). The temperature coefficient of regulator voltage test shall be performed in accordance with test method 4071 of MIL–STD–750. The device shall be temperature stabilized with current applied prior to reading regulator voltage at the specified case temperatures (see column 12 of table IV).

4.5.6 Voltage regulation ($V_{Z(reg)}$). A current at 10 percent of I_{ZM} (see column 7 of table IV) shall be maintained until thermal equilibrium is obtained and V_Z shall be noted. The current shall then be increased to a level of 50 percent of I_{ZM} and maintained at this level until thermal equilibrium is obtained, at which time the voltage change shall not exceed $V_{Z(reg)}$ (see column 8 of table IV).

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TABLE I. Group A inspection.

Inspection 1/	MIL–STD–750		Symbol	Limits 2/		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Forward voltage	4011	Condition A; I_F = 10 A dc	V_F		1.5	V dc
Reverse current (see 4.5.4)	4016	DC method V_R = column 10 of table IV;	I_{R1}		Col. 11	µA dc
Breakdown voltage (regulator voltage) (see 4.5.3)	4022	I_{ZT} = column 4 of table IV	V_Z	Col. 2	Col. 3	V dc
Subgroup 3 High temperature operation:		T_A = +150ºC				
Reverse current (see 4.5.4)	4016	DC method V_R = column 10 of table IV;	I_{R2}		Col. 13	µA dc
Subgroup 4						
Small-signal reverse breakdown impedance	4051	I_{ZT} = column 4 of table IV; I_{sig} = 10 percent of I_{ZT}	Z_Z		Col. 5	Ohm
Knee impedance	4051	I_{ZK} = 5 mA dc; I_{sig} = 10 percent of I_{ZT}	Z_{ZK}		Col. 6	Ohm
Subgroup 5 Not applicable						
Subgroup 6		JANS level only				
Surge current (see 4.5.2)	4066	Condition B; I_{ZSM} = column 9 of table IV				
End-point electrical measurements		See table I, subgroup 2 herein				
Subgroup 7						
Voltage regulation (see 4.5.6)		JANS level only; n = 22, c = 0	$V_{Z(reg)}$		Col. 8	V dc
Temperature coefficient of regulator voltage (see 4.5.5)	4071	I_{ZT} = column 4 of table IV; T_{ref} = 25ºC±3ºC; T_{test} = T_{ref} +100ºC	α_{VZ}		Col. 12	%/ºC

1/ For JANS, all devices required by the specified sampling plan shall be subjected to subgroups 2, 3, and 4 combined.
2/ Column references are from table IV herein.

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TABLE II. Groups B and C delta electrical measurements. 1/ 2/

Step	Inspection	MIL–STD–750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Forward voltage	4011	Condition A; I_F = 10 A dc	ΔV_F 3/		±50 mV dc change from previously measured value.	

1/ The delta electrical measurements for group B inspection for quality level JANS (table E–VIA of MIL–PRF–19500) are as follows:
- a. In addition to the measurements specified for subgroup 3, the measurements of step 1 of this table shall also be taken.
- b. In addition to the measurements specified for subgroup 4, the measurements of step 1 of this table shall also be taken.
- c. In addition to the measurements specified for subgroup 5, the measurements of step 1 of this table shall also be taken.

2/ The delta electrical measurements for group C inspection (table E–VII of MIL–PRF–19500) are as follows:
- a. In addition to the measurements specified for subgroup 2, the measurements of step 1 (quality level JANS only) of this table shall also be taken.
- b. In addition to the measurements specified for subgroup 3, the measurements of step 1 (quality level JANS only) of this table shall also be taken.
- c. In addition to the measurements specified for subgroup 6, the measurements of step 1 (all quality levels) of this table shall also be taken.

3/ Devices which exceed the group A limits for this test shall not be accepted.

TABLE III. Group E inspection (all quality levels) for qualification and requalification only.

Inspection 1/	MIL–STD–750		Qualification inspection
	Method	Conditions	
Subgroup 1			n = 45, c = 0
Thermal shock	1056		
Hermetic seal	1071		
End-point electrical measurements		See table I, subgroup 2 herein.	
Subgroup 2			n = 45, c = 0
Intermittent operation life	1037	6,000 cycles. See 4.5.1.	
End-point electrical measurements		See table I, subgroup 2 herein.	
Subgroup 4			
Thermal impedance curves		See MIL–PRF–19500.	
Subgroup 5 and 6			
Not applicable			

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TABLE IV. Characteristics and test ratings.

Device type	Col 1	Col 2	Col 3	Col 4	Col 5	Col 6	Col 7	Col 8	Col 9	Col 10	Col 11	Col 12	Col 13
	V_Z Nom	V_Z Min	V_Z Max	I_{ZT} 1/	Z_Z	Z_K	I_{ZM} 1/	$V_{Z(reg)}$	I_{ZSM} 1/	V_R	I_{R1}	α_{VZ}	I_{R2} T_A = 150ºC
	V dc	V dc	V dc	mA dc	ohms	ohms	mA dc	V dc	A dc	V dc	µA dc	%/°C	µA dc
1N4549	3.9	3.71	4.09	3,200	0.16	400	12,400	0.66	40.0	0.5	150	–.06	2/
1N4550	4.3	4.09	4.51	2,900	0.16	500	11,050	0.58	38.0	0.5	150	–.05	2/
1N4551	4.7	4.47	4.93	2,650	0.12	600	10,100	0.40	35.0	1.0	100	±.025	2/
1N4552	5.1	4.85	5.35	2,450	0.12	650	9,300	0.36	32.0	1.0	20	±.030	2/
1N4553	5.6	5.32	5.88	2,250	0.12	900	8,500	0.34	30.0	1.0	20	+.050	2/
1N4554	6.2	5.89	6.51	2,000	0.14	1,000	7,650	0.36	25.0	2.0	20	+.055	2/
1N3305	6.8	6.46	7.14	1,850	0.2	750	7,000	0.4	37.0	4.5	300	+.057	1,000
1N3306	7.5	7.13	7.87	1,700	0.3	350	6,360	0.5	33.0	5.0	125	+.067	750
1N3307	8.2	7.79	8.61	1,500	0.4	250	5,800	0.6	29.0	5.4	50	+.070	500
1N3308	9.1	8.65	9.55	1,370	0.5	250	5,240	0.7	26.5	6.1	25	+.075	400
1N3309	10	9.50	10.50	1,200	0.6	250	4,760	0.9	24.0	6.7	25	+.081	300
1N3310	11	10.45	11.55	1,100	0.8	250	4,330	1.0	21.5	8.4	10	+.085	200
1N3311	12	11.40	12.60	1,000	1.0	250	3,970	1.1	20.0	9.1	10	+.079	200
1N3312	13	12.35	13.65	960	1.1	250	3,750	1.2	18.5	9.9	10	+.080	200
1N3313	14	13.30	14.70	890	1.2	250	3000	1.4	17.0	11.4	10	+.080	200
1N3314	15	14.25	15.75	830	1.4	250	3,170	1.5	15.5	11.4	10	+.082	200
1N3315	16	15.20	16.80	780	1.6	250	2,970	1.6	14.75	12.2	10	+.083	200
1N3316	17	16.15	17.85	740	1.8	250	2,500	1.8	13.75	13.0	10	+.085	200
1N3317	18	17.10	18.90	700	2.0	250	2,640	1.9	12.75	13.7	10	+.085	200
1N3318	19	18.05	19.95	660	2.2	250	2,200	2.0	12.75	13.7	10	+.086	200
1N3319	20	19.00	21.00	630	2.4	250	2,380	2.3	11.75	15.2	10	+.086	200
1N3320	22	20.90	23.10	570	2.5	250	2,160	2.5	10.5	16.7	10	+.087	200
1N3321	24	22.80	25.20	520	2.6	250	1,980	2.6	9.75	18.2	10	+.088	200
1N3322	25	23.75	26.25	500	2.7	250	1550	2.8	9.00	18.2	10	+.089	200
1N3323	27	25.65	28.35	460	2.8	250	1,760	2.9	8.25	20.6	10	+.090	200
1N3324	30	28.50	31.50	420	3.0	300	1,590	3.0	7.75	22.8	10	+.091	200
1N3325	33	31.35	34.65	380	3.2	300	1,440	3.2	7.25	25.1	10	+.092	200

See footnotes at end of table.

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TABLE IV. Characteristics and test ratings – Continued.

Device type	Col 1	Col 2	Col 3	Col 4	Col 5	Col 6	Col 7	Col 8	Col 9	Col 10	Col 11	Col 12	Col 13
	V_Z Nom	V_Z Min	V_Z Max	I_{ZT} 1/	Z_Z	Z_{ZK}	I_{ZM} 1/	$V_{Z(reg)}$	I_{ZSM} 1/	V_R	I_{R1}	α_{VZ}	I_{R2} T_A = 150°C
	V dc	V dc	V dc	mA dc	ohms	ohms	mA dc	V dc	A dc	V dc	µA dc	%/°C	µA dc
1N3326	36	34.20	37.80	350	3.5	300	1,320	3.4	6.5	27.4	10	+.093	200
1N3327	39	37.10	40.90	320	4.0	350	1,220	3.6	5.88	29.7	10	+.094	200
1N3328	43	40.90	45.10	290	4.5	400	1,110	3.8	5.38	32.7	10	+.095	200
1N3329	45	42.75	47.25	280	4.5	400	930	3.9	5.14	32.7	10	+.095	200
1N3330	47	44.65	49.35	270	5.0	400	1,020	4.0	4.90	35.8	10	+.095	200
1N3331	50	47.50	52.50	250	5.0	500	830	4.2	4.76	38.8	10	+.096	200
1N3332	51	48.45	53.55	245	5.2	500	930	4.4	4.36	38.8	10	+.096	200
1N3333	52	49.40	54.60	240	5.5	500	790	4.65	4.38	42.6	10	+.096	200
1N3334	56	53.20	58.80	220	6	500	850	4.75	4.13	42.6	10	+.096	200
1N3335	62	58.90	65.10	200	7	600	770	5.0	3.68	47.1	10	+.097	200
1N3336	68	64.60	71.40	180	8	600	700	5.5	3.35	51.7	10	+.097	200
1N3337	75	71.25	78.75	170	9	600	640	5.75	3.00	56.0	10	+.098	200
1N3338	82	77.90	86.10	150	11	700	580	6.25	2.75	62.2	10	+.098	200
1N3339	91	86.45	95.55	140	15	800	530	6.75	2.35	69.2	10	+.099	200
1N3340	100	95.00	105.0	120	20	900	480	7.5	2.25	76.0	10	+.100	200
1N3341	105	99.75	110.25	120	25	1000	380	8.25	2.15	83.0	10	+.100	200
1N3342	110	104.50	115.5	110	30	1100	430	9.0	2.05	83.0	10	+.100	200
1N3343	120	114.00	126.0	100	40	1200	400	9.5	1.88	91.2	10	+.100	200
1N3344	130	123.50	136.5	95	50	1300	370	10.0	1.73	98.8	10	+.100	200
1N3345	140	133.00	147.0	90	60	1400	325	11.00	1.61	114.0	10	+.100	200
1N3346	150	142.50	157.5	85	75	1500	320	12.0	1.50	114.0	10	+.100	200
1N3347	160	152.00	168.0	80	80	1600	300	13.0	1.43	121.6	10	+.100	200
1N3348	175	166.25	183.75	70	85	1700	230	13.75	1.34	121.6	10	+.100	200
1N3349	180	171.00	189.0	68	90	1800	260	14.5	1.25	136.8	10	+.100	200
1N3350	200	190.00	210.0	65	100	2000	240	16.0	1.10	152.0	10	+.100	200

1/ Unless otherwise specified herein, all inspections shall be made at T_C of 25°C ±3°C.
2/ This test is not applicable for devices 1N4549B, RB through 1N4554B, RB.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL–PRF–19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

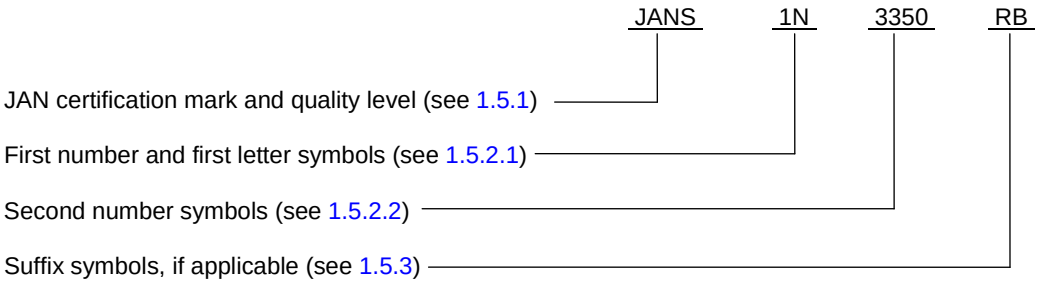
b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. The complete PIN, see 1.5 and 6.4.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 PIN construction example. The PINs for encapsulated devices are constructed using the following form.



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6.5 List of PINs. The following is a list of possible PINs available for devices covered by this specification sheet. The reverse polarity devices, suffix RB PINs, are not listed below. For reverse polarity devices substitute "RB" for "B".

JAN1N4549B	JANTX1N4549B	JANTXV1N4549B	JANS1N4549B
JAN1N4550B	JANTX1N4550B	JANTXV1N4550B	JANS1N4550B
JAN1N4551B	JANTX1N4551B	JANTXV1N4551B	JANS1N4551B
JAN1N4552B	JANTX1N4552B	JANTXV1N4552B	JANS1N4552B
JAN1N4553B	JANTX1N4553B	JANTXV1N4553B	JANS1N4553B
JAN1N4554B	JANTX1N4554B	JANTXV1N4554B	JANS1N4554B
JAN1N3305B	JANTX1N3305B	JANTXV1N3305B	JANS1N3305B
JAN1N3306B	JANTX1N3306B	JANTXV1N3306B	JANS1N3306B
JAN1N3307B	JANTX1N3307B	JANTXV1N3307B	JANS1N3307B
JAN1N3308B	JANTX1N3308B	JANTXV1N3308B	JANS1N3308B
JAN1N3309B	JANTX1N3309B	JANTXV1N3309B	JANS1N3309B
JAN1N3310B	JANTX1N3310B	JANTXV1N3310B	JANS1N3310B
JAN1N3311B	JANTX1N3311B	JANTXV1N3311B	JANS1N3311B
JAN1N3312B	JANTX1N3312B	JANTXV1N3312B	JANS1N3312B
JAN1N3313B	JANTX1N3313B	JANTXV1N3313B	JANS1N3313B
JAN1N3314B	JANTX1N3314B	JANTXV1N3314B	JANS1N3314B
JAN1N3315B	JANTX1N3315B	JANTXV1N3315B	JANS1N3315B
JAN1N3316B	JANTX1N3316B	JANTXV1N3316B	JANS1N3316B
JAN1N3317B	JANTX1N3317B	JANTXV1N3317B	JANS1N3317B
JAN1N3318B	JANTX1N3318B	JANTXV1N3318B	JANS1N3318B
JAN1N3319B	JANTX1N3319B	JANTXV1N3319B	JANS1N3319B
JAN1N3320B	JANTX1N3320B	JANTXV1N3320B	JANS1N3320B
JAN1N3321B	JANTX1N3321B	JANTXV1N3321B	JANS1N3321B
JAN1N3322B	JANTX1N3322B	JANTXV1N3322B	JANS1N3322B
JAN1N3323B	JANTX1N3323B	JANTXV1N3323B	JANS1N3323B
JAN1N3324B	JANTX1N3324B	JANTXV1N3324B	JANS1N3324B
JAN1N3325B	JANTX1N3325B	JANTXV1N3325B	JANS1N3325B
JAN1N3326B	JANTX1N3326B	JANTXV1N3326B	JANS1N3326B
JAN1N3327B	JANTX1N3327B	JANTXV1N3327B	JANS1N3327B
JAN1N3328B	JANTX1N3328B	JANTXV1N3328B	JANS1N3328B
JAN1N3329B	JANTX1N3329B	JANTXV1N3329B	JANS1N3329B
JAN1N3330B	JANTX1N3330B	JANTXV1N3330B	JANS1N3330B
JAN1N3331B	JANTX1N3331B	JANTXV1N3331B	JANS1N3331B
JAN1N3332B	JANTX1N3332B	JANTXV1N3332B	JANS1N3332B
JAN1N3333B	JANTX1N3333B	JANTXV1N3333B	JANS1N3333B
JAN1N3334B	JANTX1N3334B	JANTXV1N3334B	JANS1N3334B
JAN1N3335B	JANTX1N3335B	JANTXV1N3335B	JANS1N3335B
JAN1N3336B	JANTX1N3336B	JANTXV1N3336B	JANS1N3336B
JAN1N3337B	JANTX1N3337B	JANTXV1N3337B	JANS1N3337B
JAN1N3338B	JANTX1N3338B	JANTXV1N3338B	JANS1N3338B
JAN1N3339B	JANTX1N3339B	JANTXV1N3339B	JANS1N3339B
JAN1N3340B	JANTX1N3340B	JANTXV1N3340B	JANS1N3340B
JAN1N3341B	JANTX1N3341B	JANTXV1N3341B	JANS1N3341B
JAN1N3342B	JANTX1N3342B	JANTXV1N3342B	JANS1N3342B
JAN1N3343B	JANTX1N3343B	JANTXV1N3343B	JANS1N3343B
JAN1N3344B	JANTX1N3344B	JANTXV1N3344B	JANS1N3344B
JAN1N3345B	JANTX1N3345B	JANTXV1N3345B	JANS1N3345B
JAN1N3346B	JANTX1N3346B	JANTXV1N3346B	JANS1N3346B
JAN1N3347B	JANTX1N3347B	JANTXV1N3347B	JANS1N3347B
JAN1N3348B	JANTX1N3348B	JANTXV1N3348B	JANS1N3348B
JAN1N3349B	JANTX1N3349B	JANTXV1N3349B	JANS1N3349B
JAN1N3350B	JANTX1N3350B	JANTXV1N3350B	JANS1N3350B

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6.6 Supersession information and non-included numerical types.

6.6.1 Re-identification of previous duplicate currents I_Z to currents I_{ZM} and I_{ZT}. Previous revisions of this specification used the symbol I_Z in two different columns of table IV. In MIL-PRF-19500/358E (dated 23 July 1999) and earlier revisions, current I_Z was used in columns 4 and 7. These were re-identified as I_{ZT} (column 4) and I_{ZM} (column 7) with the issuance of MIL–PRF–19500/358F; dated 12 July 2013.

6.6.1 Re-identification of the package outline. Previous revisions of this specification identified the device package as a DO–5 diode outline. This issue of the specification has adjusted the dimensions on figure 1 and the description of the package as supplied to a modified DO–5 base with a DO–4 cap.

6.6.2 Numerical diode-types not covered by past revisions. The 10 numerical types listed below were not covered in the specification sheet until they were included in the issuance of MIL–S–19500/358D, dated 26 May 1994. These previous non-included types (all from the 1N33XX series) have intermediate voltage-characteristics, etc., parameter deemed (as overlapping and) that were considered not requisite for military-equipment circuit applications. These 10 numerical types were then removed with the issuance of MIL–PRF–19500/358F; dated 12 July 2013.

Previous non-included numerical types were as follows: 3313, 3316, 3318, 3322, 3329, 3331, 3333, 3341, 3345, and 3348.

6.7 Voltage-range sequence factors for diodes covered herein. The 1N4549 through 1N4555 (both B and RB) series of diodes are designed for applications at a lower range of nominal-voltage levels than the 1N3305 through 1N3350 series of diodes.

6.8 Symbols used in this specification sheet. The following symbols are used in this specification sheet. The definition associated with the symbol shall be as defined in MIL–PRF–19500 or herein.

I_F	Forward current, DC value, no alternating component.
I_{R1}	Reverse current.
I_{R3}	Reverse current, dc max (after life test).
I_{sig}	Detector signal current.
I_{ZSM}	Maximum zener surge current.
I_Z	Maximum dc current.
I_{ZK}	Regulator or reference current, dc near breakdown knee.
I_{ZM}	Regulator or reference current, dc maximum rated current.
I_{ZT}	Zener test current.
P_T	Total power dissipation, all terminals.
$R_{\theta JC}$	Thermal resistance, junction to case.
T_A	Ambient or free air temperature.
T_C	Case temperature.
T_J	Junction temperature.
T_{STG}	Storage temperature.
V_F	Forward voltage.
V_R	Reverse voltage.
V_Z	Regulator voltage.
$V_{Z(reg)}$	Voltage regulation.
Z_Z	Regulator impedance.
Z_{ZK}	Knee impedance.
α_{vz}	Temperature coefficient of regulator voltage.

6.9 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at "Semiconductor@.dla.mil" or by facsimile (614) 693–1642 or DSN 850–6939.

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6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – CR
Air Force – 85
NASA – NA
DLA – CC

Review activities:
Army – AR, MR, SM
Air Force – 99

Preparing activity:
DLA – CC

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